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Advancing Precision Livestock Farming: Robust Country Chicken Detection via FeatherNet Fusion-YOLO and HenSense

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INTRODUCTION

Vision-based systems enable scalable, non-invasive monitoring in precision poultry farming, helping farmers efficiently manage flocks, improve animal welfare, and operate reliably in real farm environments.

PROBLEM STATEMENT

- Most existing vision-based poultry systems are designed and tested in controlled indoor environments. When applied to free-range rural farms, their performance drops significantly.

- In real farms, chickens often move in dense groups, causing heavy occlusion. Lighting conditions vary strongly between day and night, native breeds show large appearance differences, and rural backgrounds are cluttered and uncontrolled. These factors make reliable detection very challenging.

WHY EXISTING POULTRY DATASETS FAIL

- Most publicly available poultry datasets are collected in **controlled indoor farm environments** with clean backgrounds, uniform lighting, and low crowd density, typically focusing on **commercial broiler breeds** with only a few birds per image.

- These datasets rarely include **night-time scenes or challenging lighting conditions**, causing models trained on them to perform well in laboratory settings but **fail to generalize in real free-range rural farms**.

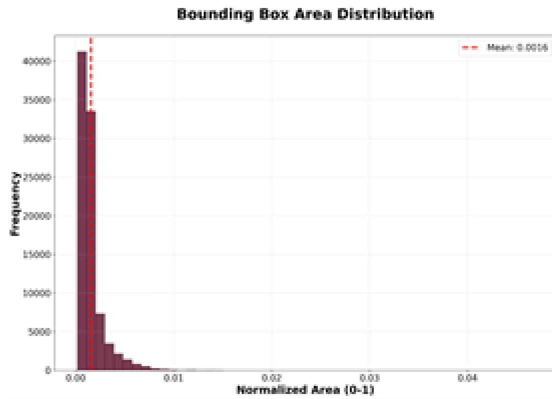
Dataset	No. Images train	val	test	total	Real Farm	Breed Diversity	Night Conditions	Max. Img. Resolution	Total Instances	Avg per Image
Disease Detection & Health Assessment										
Tanzania Poultry [10]	4,768	1,362	682	6,812	✓	✓	×	Variable	—	—
Tanzania PCR-Lab [10]	—	—	—	1,255	✓	✓	×	Variable	—	—
Lithuania Dropping [11]	—	—	—	1,800	✓	×	×	Variable	—	—
Sick Broiler (Zhuang) [12]	—	—	—	14,728	✓	×	×	Variable	—	—
Elmessery Thermal [13]	—	—	—	10,000	✓	×	×	Variable	50,000	5.0
Roboflow Sick [14]	167	21	21	209	✓	×	×	Variable	—	—
Behavior Monitoring & Tracking										
UGA Broiler [15]	8,000	2,000	200	10,200	×	×	×	1440×1080	—	—
ChickTrack [16]	— Video dataset —			—	✓	✓	×	HD Video	—	—
Thailand YOLOv11 [17]	—	—	—	1,716	✓	×	×	Variable	—	—
Trajectory Cluster [18]	—	—	—	—	✓	×	×	Variable	—	—
Individual Identification										
Chicks4FreeID [19]	800	100	100	1,000	✓	✓	×	Variable	1,270	1.27
Roboflow Chicken [20]	148	19	18	185	✓	✓	×	Variable	—	—
HenSense (Ours)	**11,196**	—	**2,799**	**13,995**	✓	✓	✓	**Multiple**	**195,163**	**13.95**

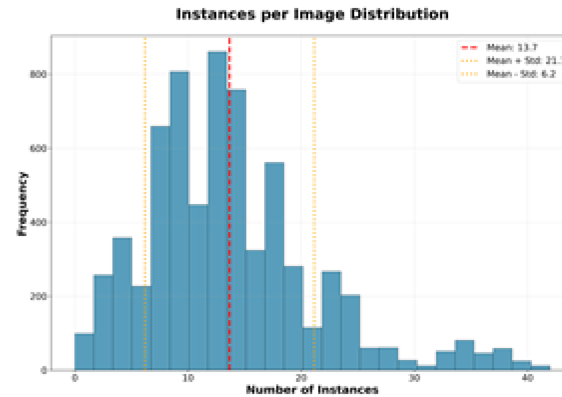
HENSENSE DATASET

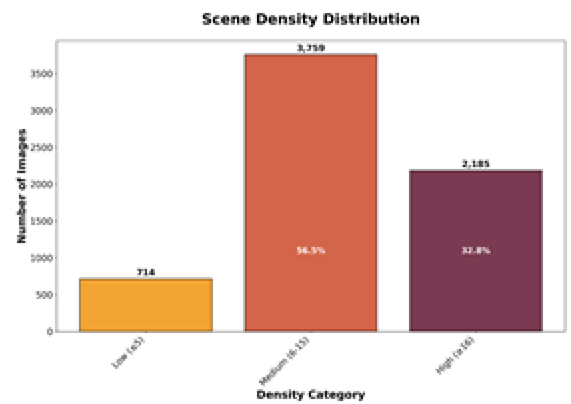
- To address the limitations of existing datasets, we introduce HenSense, a large-scale dataset collected from real free-range rural poultry farms.
- HenSense contains images captured under diverse conditions, including dense flocks, day and night scenes, and different native chicken breeds. The dataset is designed to reflect real-world farm environments.
- This dataset enables realistic benchmarking and development of robust poultry detection models.

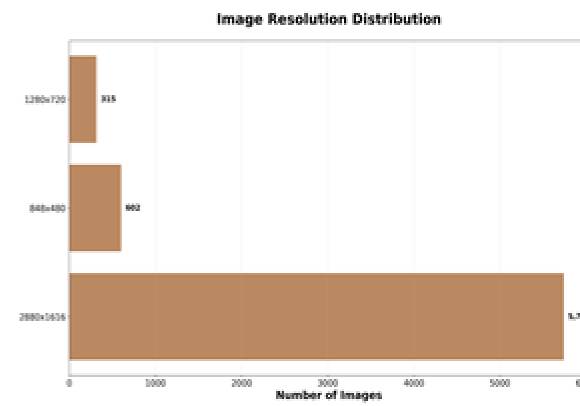
DATASET STATISTICS

- Total images: 13,995 high-resolution RGB images
- Total annotations: 195,163 bounding boxes
- Average density: ~14 chickens per image
- Farm setup: Real free-range rural poultry farms
- Data conditions: Day and night scenes included
- Breed diversity: Multiple native chicken breeds
- Viewpoints: Different camera angles and farm layouts
- Scene complexity: Dense flocks, occlusion, and cluttered backgrounds



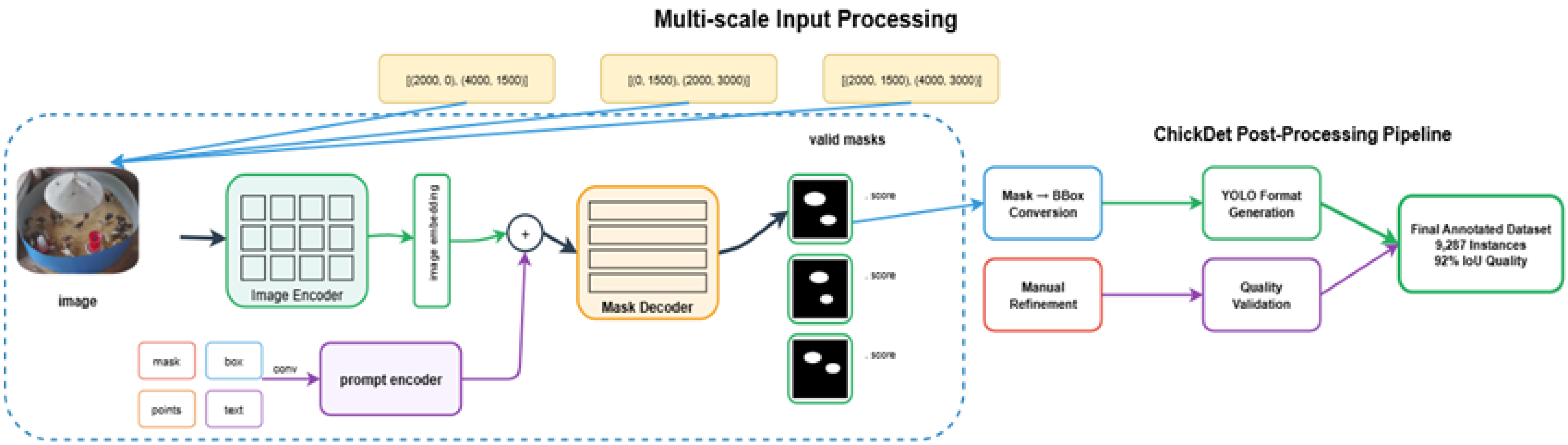






DATASET VISUAL DIVERSITY

ANNOTATION PIPELINE



- First, each poultry image is given as input to the system. To handle chickens of different sizes and crowd levels, the image is processed at multiple scales.
- Next, LangSAM is used to automatically generate segmentation masks for the chickens in the image. These masks highlight the regions where chickens are present.
- Then, the generated masks are converted into bounding boxes, which are required for training object detection models.
- After that, a manual refinement step is performed. In this step, humans check the annotations and correct any missing or incorrect boxes.
- Finally, the verified annotations are converted into YOLO format, and the final annotated dataset is created for training and evaluation.

PROPOSED METHODOLOGY

Step 1: Dataset Preparation

- Construct the HenSense dataset from real free-range farm environments
- Apply a semi-automated LangSAM-based annotation pipeline
- Validate annotations through manual refinement and quality checks

Step 2: Baseline Evaluation

- Benchmark multiple lightweight YOLO variants on HenSense
- Evaluate robustness under dense flocks, occlusion, and lighting variations

Step 3: Model Design

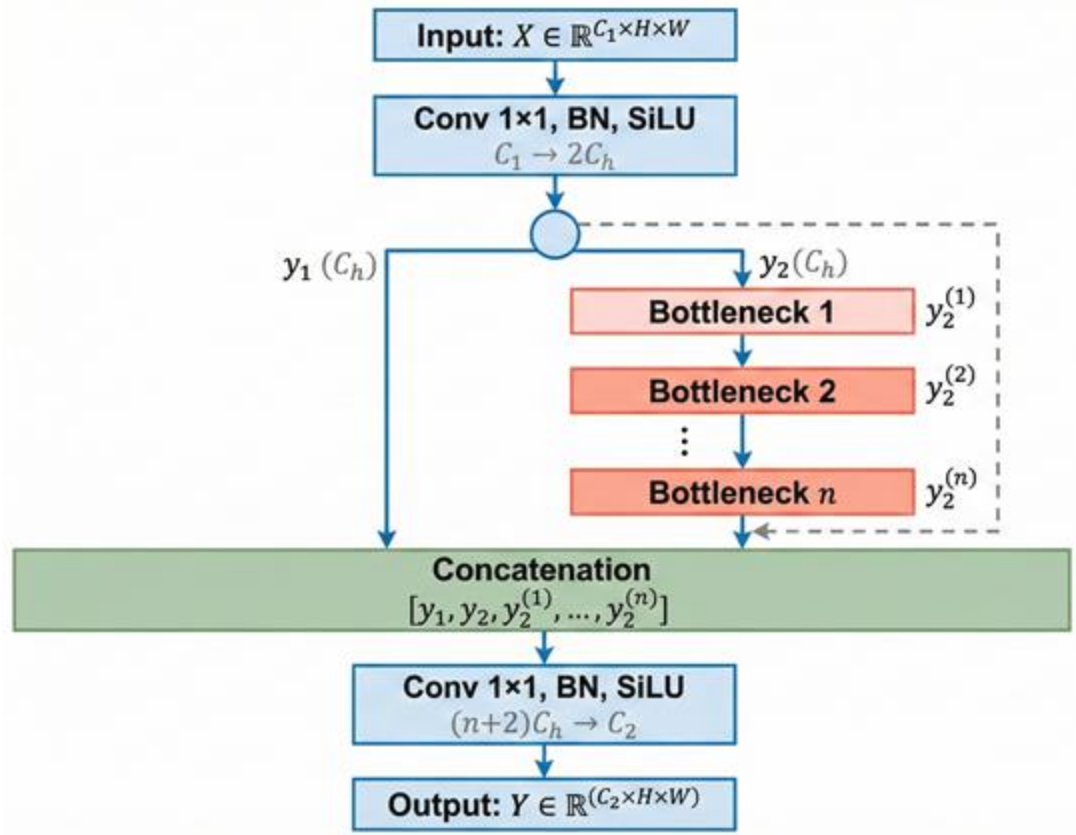
- Analyze baseline failures and performance bottlenecks
- Design FeatherNet Fusion-YOLO with multi-path feature fusion
- Focus on robustness and computational efficiency

Step 4: Performance Analysis

- Evaluate detection accuracy using mAP metrics
- Analyze efficiency using parameters and GFLOPs
- Compare against existing lightweight detectors

FEATHERNET FUSION-YOLO

- Multi-path feature fusion for dense and occluded scenes
- Bottleneck-based feature refinement
- Combines shallow and deep features through concatenation
- Lightweight design for efficient deployment



EXPERIMENTAL SETUP

Dataset

- 13,995 images, 195,163 annotations
- Train/Test split: 80% / 20%
- Avg density: 13.95 birds/image

Training

- Ultralytics YOLO framework
- 100 epochs, AdamW optimizer
- Batch size: 16–64
- Cosine LR scheduler

Evaluation

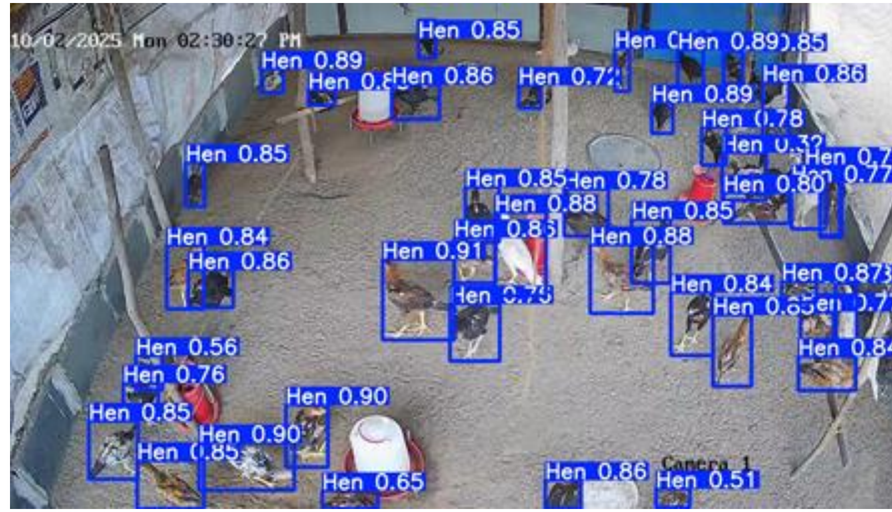
- Precision, Recall
- mAP@0.5, mAP@0.5:0.95
- Parameters & GFLOPs

EXPERIMENTAL RESULTS

Table 2. Lightweight YOLO models and FeatherNet Fusion-YOLO on HenSense. Models with low parameter count and FLOPs suitable for edge deployment.

Model	Params	FLOPs	P	R	mAP50	mAP50-95
YOLOv5 Series [22]						
YOLOv5n	2.50	7.1	0.863	0.872	0.919	0.713
YOLOv8 Series [23]						
YOLOv8n	3.01	8.1	0.870	0.874	0.923	0.722
YOLOv9 Series						
YOLOv9t	2.97	7.6	0.861	0.877	0.926	0.720
YOLOv10 Series [24]						
YOLOv10n	2.26	6.5	0.860	0.871	0.920	0.725
YOLOv12 Series [26]						
YOLOv12n	2.56	6.3	0.870	0.876	0.924	0.724
Attention Model						
FeatherNet	**2.14**	**6.1**	**0.872**	**0.878**	**0.926**	**0.723**

EXPERIMENTAL RESULTS



DISCUSSIONS

Interpretation of Results

- FeatherNet Fusion-YOLO achieves competitive accuracy with significantly fewer parameters and FLOPs
- Multi-path feature fusion improves detection in dense and occluded poultry scenes
- Performance remains stable across challenging lighting and background conditions

Strengths

- Real-world dataset collected from free-range rural farms
- Robust detection under dense flocks and heavy occlusion
- Lightweight design suitable for edge and on-farm deployment
- Efficient annotation pipeline reduces manual labeling effort

Limitations

- Evaluation limited to still images (no temporal information)
- Extreme night-time and severe occlusion cases remain challenging

APPLICATIONS / USE CASES

- **Automated Poultry Counting:** Accurate bird counting in free-range and rural farm environments
- **Welfare Monitoring:** Continuous observation of flock behavior and movement patterns
- **Farm Management:** Support decision-making for feeding, housing, and space utilization
- **Smallholder Farms:** Low-cost, edge-deployable solutions for rural poultry farmers
- **Precision Livestock Farming:** Scalable vision-based monitoring under real-world conditions

CONCLUSION & FUTURE WORK

Contributions:

– Introduced **HenSense**, a large-scale real-world dataset for free-range poultry detection
– Proposed **FeatherNet Fusion-YOLO**, a lightweight and robust detection model
– Addressed dense flocks, occlusion, lighting variation, and breed diversity
– Designed for efficient real-world and edge deployment

Research Outcomes:

– Competitive detection accuracy in challenging farm environments
– Lower parameter count and FLOPs compared to lightweight YOLO baselines
– Improved robustness under dense and occluded scenes
– Practical applicability for rural and smallholder farms

Future Work:

– Expand the dataset to more geographic regions and farm types
– Explore multi-modal data (thermal, depth, environmental sensors)
– Evaluate long-term real-time deployment on edge devices

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THANK YOU